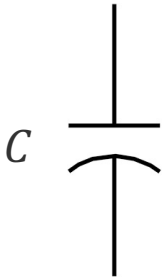
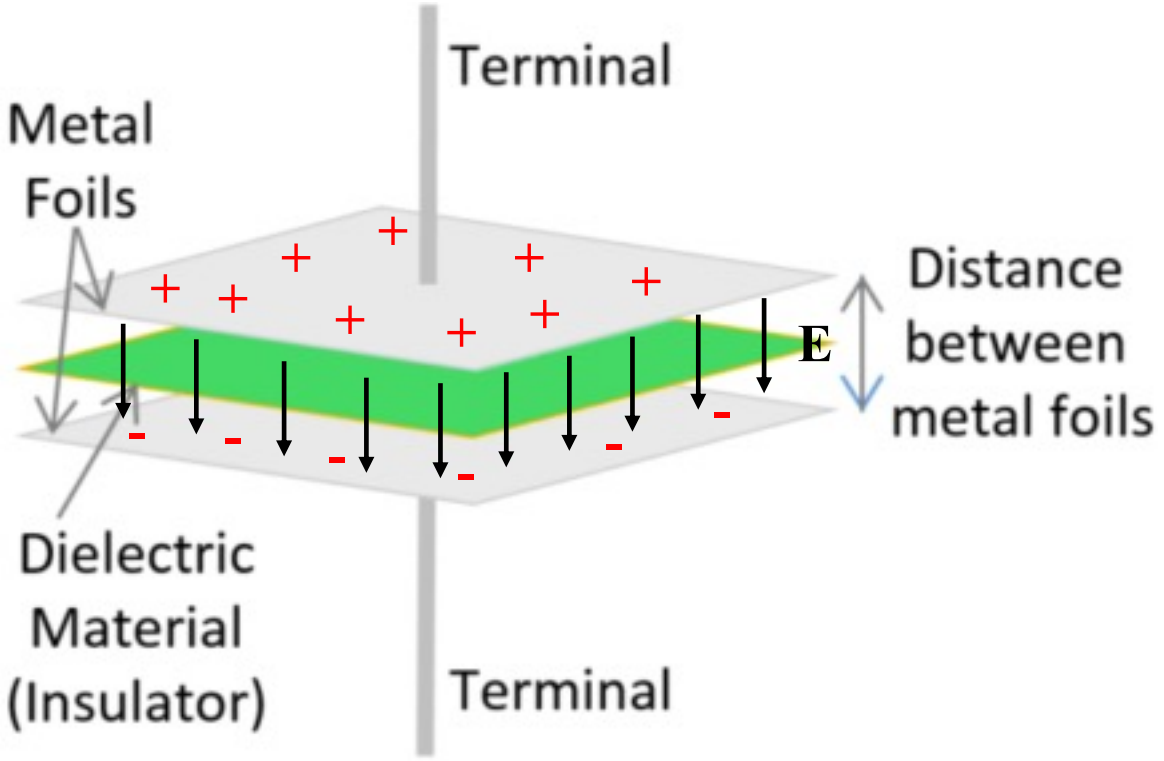
Chapter 25: Capacitance

The following subjects will be discussed

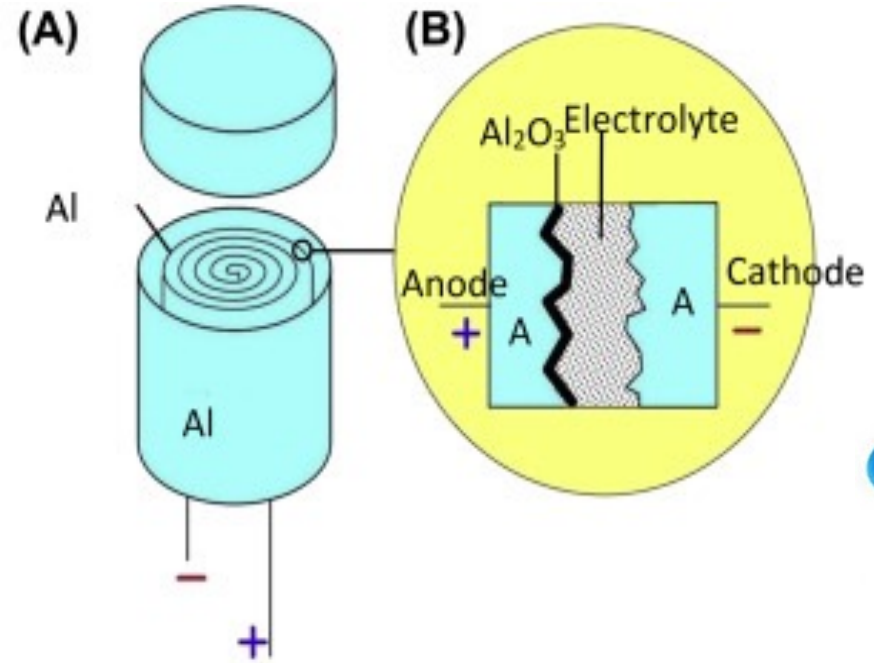
- Definition of Capacitance.
- Capacitance calculations of several structures.
- Energy stored in capacitors.
- Capacitors in electric circuits.
- Examples.

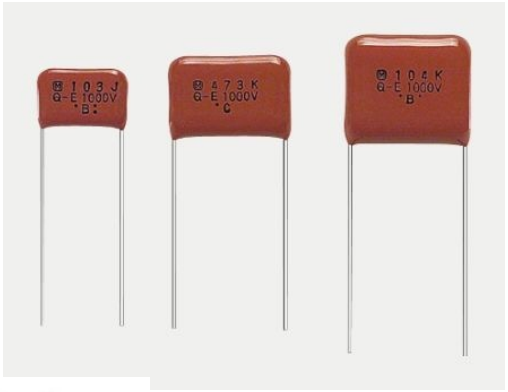
Capacitors


C


Terminal
Metal
Foils
E
Distance
between
metal foils
Dielectric
Material
(Insulator)
Terminal

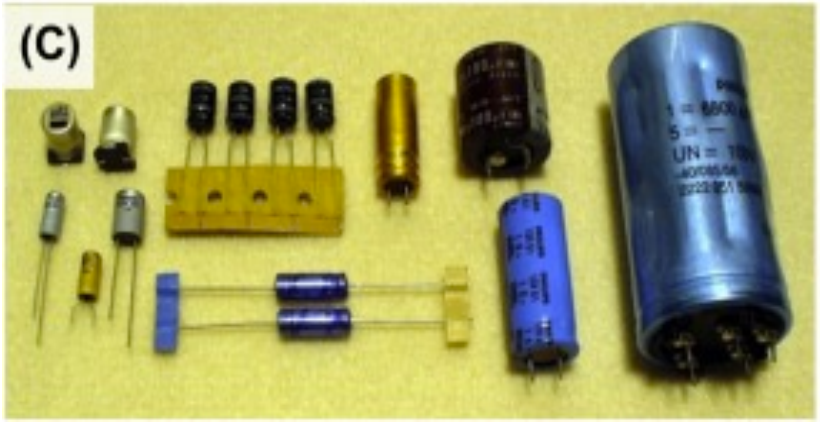
Capacitors













Capacitors

We can find capacitors in almost every electronic circuit; analog or digital. They serve many purposes, such as:

Temporary storage of charge/energy.

Timing

Coupling circuits together to block DC/allow AC

Filtering and signal processing

Definition of Capacitance

The amount of charge that can be stored for every 1 volt of potential difference, is called Capacitance.

$$C = \frac{Q}{V}$$ Unit is Farad (F)

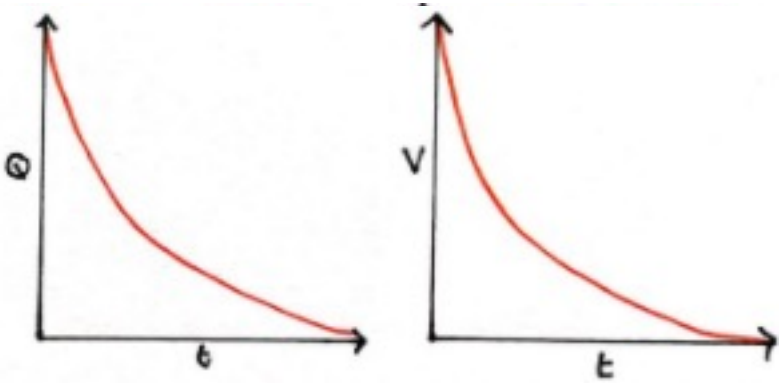
Capacitance does not mean capacity.
A capacitor is not a tank!

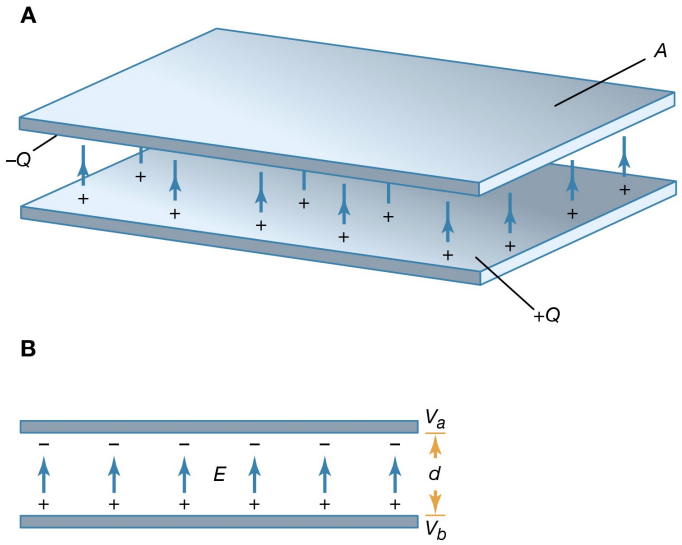
Depends on (1) shape (2) dimensions (3) filling material.

Example: The parallel-plate capacitor

$$C = \varepsilon_o \frac{A}{d}$$

Discharging





Parallel-plate Capacitor Example

Consider a parallel-plate capacitor with plates of dimensions 2 cm x 2 cm and separation of 1 mm. The filling dielectric is air.

Calculate

1) The capacitance of this structure.

$$C = \varepsilon_o \frac{A}{d} = 8.85\times10^{-12}\,\frac{0.02\times0.02}{0.001} = 3.54\ \text{pF}$$

2) The maximum charge it can handle. $Q = C\,V$ ➧ $Q_{max} = C\,V_{max} = C\,E_{max}\,d$

$$Q_{max} = 3.54\times10^{-12}\times0.001\times3\times10^{6} = 0.0106\ \mu\text{C}$$

3) Repeat for paper filling (ε_r=3.7).

$$C = \varepsilon_o \varepsilon_r \frac{A}{d} = 8.85\times10^{-12}\times3.7\,\frac{0.02\times0.02}{0.001} = 13.1\ \text{pF}$$

$$Q_{max} = 13.1\times10^{-12}\times0.001\times15\times10^{6} = 0.20\ \mu\text{C}$$

20 Times more!!

Material	E_{cr} (MV/m)	Material	E_{cr} (MV/m)
Air (atmospheric pressure)	3	Bakelite	25
Barium titanate ($BaTiO_3$)	7.5	Glass (plate)	30
Freon	~ 8	Paraffin	~ 30
Germanium	~ 10	Silicon (Si)	~ 30
Wood (douglas fir)	~ 10	Alumina	~ 35
Porcelain	11	Gallium arsenide	~ 40
Oil (mineral)	15	Polyethylene	47
Paper (impregnated)	15	Mica	200
Polystyrene	20	Fused quartz (SiO_2)	~ 1000
Teflon	20	Silicon nitride (Si_3N_4)	~ 1000
Rubber (hard)	25	Vacuum	∞

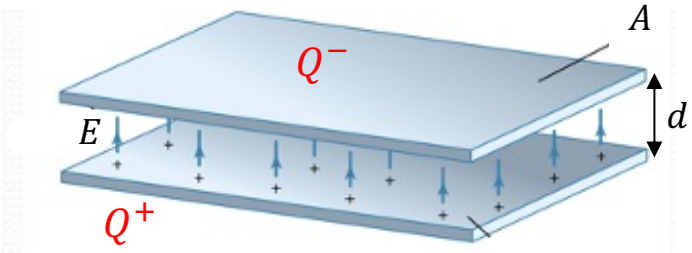
From week 3

Capacitance Calculations: Parallel-plate Capacitor

To find the capacitance of any given structure, the procedure is as follows.

1) Identify the two surface between which the capacitance is to be found.
2) Assume Q^+ and Q^- on the two surfaces.
3) Find the corresponding electric field E.
4) Find the potential difference using $V = -\int \vec{E} \cdot \overrightarrow{dl}$.
5) Use $C = Q/V$.

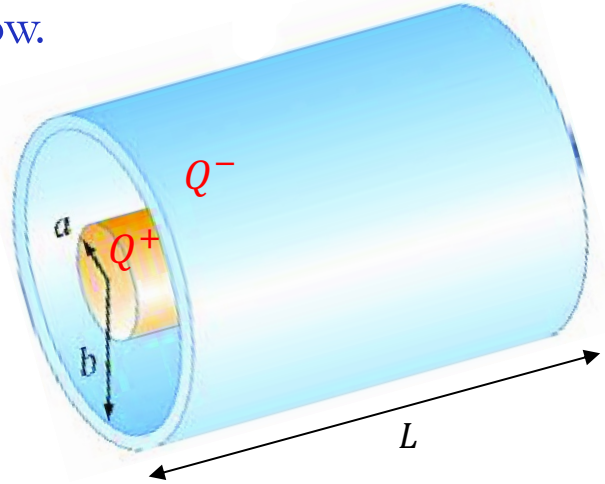
Let's apply it to the parallel-plate structure with $d \ll$ other dimensions (the plates can be considered infinite)



1) Top and bottom.
2) Assume Q^+ and Q^- on the two surfaces.
3) The electric field due to bottom plate is $\vec{E} = \frac{Q^+}{2\varepsilon_o A}\vec{a}_n$, top plate $\vec{E} = \frac{-Q^-}{2\varepsilon_o A}(-\vec{a}_n)$, total field between the plates $\vec{E} = \frac{Q}{\varepsilon_o A}\vec{a}_n$.
4) The potential is obtained by integrating $\vec{E}$ from one plate to the other $V = \frac{Q\,d}{\varepsilon_o A}$.
5) Use $C = \frac{Q}{V} = \frac{\varepsilon_o\, A}{d}$.

Capacitance Calculations: Cylindrical Capacitor

Let's apply the procedure to the cylindrical structure below.

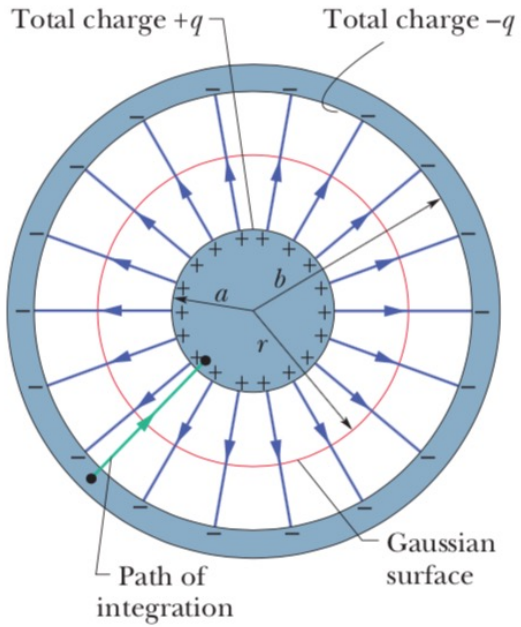


The electric field in the region $a \le r \le b$ using Gauss' Law.

The charge enclosed $= Q$

The surface area $= 2\pi rL$

So, the field inside the cylinder $\vec{E} = \frac{Q}{2\pi\varepsilon_o rL}\vec{a}_r$



The potential $V = -\int \vec{E} \cdot \vec{dl}$. Integrate from a to b.

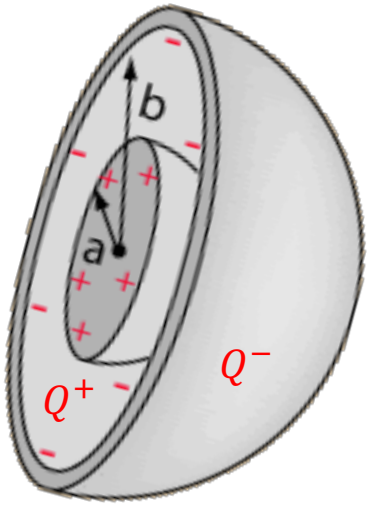
$$V = \frac{-Q}{2\pi\varepsilon_o L}\int_a^b \frac{1}{r}\,dr \quad \Rightarrow \quad V = \frac{Q}{2\pi\varepsilon_o L}\ln(b/a)$$

Now. $C = \frac{Q}{V}$.

$$C = \frac{2\pi\varepsilon_o L}{\ln(b/a)}$$

Capacitance Calculations: Spherical Capacitor

Let's apply the procedure to the spherical structure below.



The electric field in the region $a \le r \le b$ using Gauss' Law.

The charge enclosed $= Q$

The surface area $= 4\pi r^2$

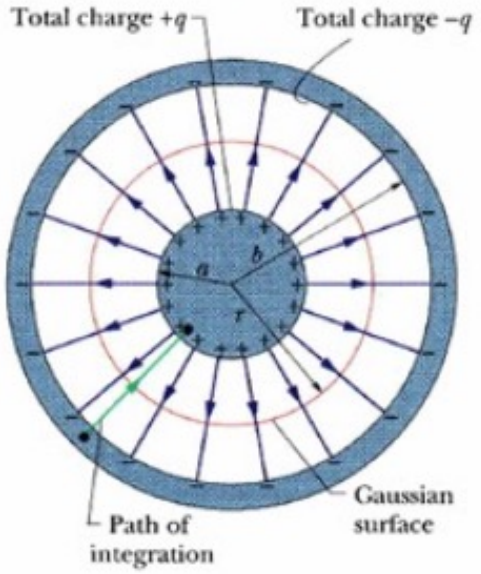
So, the field inside the cylinder $\vec{E} = \frac{Q}{4\pi\varepsilon_o r^2}\vec{a}_r$

The potential $V = -\int \vec{E} \cdot \vec{dl}$. Integrate from a to b.

$$V = \frac{-Q}{4\pi\varepsilon_o}\int_a^b \frac{1}{r^2}\,dr \quad \Rightarrow \quad V = \frac{Q}{4\pi\varepsilon_o}\left(\frac{1}{a} - \frac{1}{b}\right)$$

Now. $C = \frac{Q}{V}$.

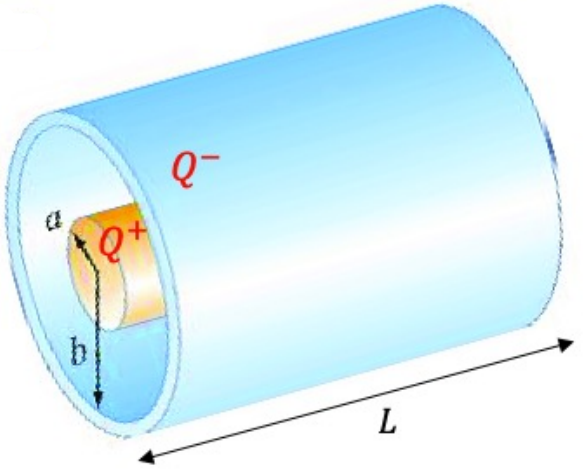
$$C = 4\pi\varepsilon_o \frac{ab}{b-a}$$



Example: A Design Consideration

Consider a cylindrical capacitor with a, $b = 2a$ and L.

$$C = \frac{2\pi\varepsilon_o L}{\ln(b/a)}$$



To increase the capacitance, which is more efficient 10% change in a or L?

10% in change in a $\quad C_1 = \frac{2\pi\varepsilon_o L}{\ln(2/1.1)}$

10% in change in L $\quad C_2 = \frac{2.2\,\pi\varepsilon_o L}{\ln(2)}$

$$\frac{C_1}{C_2} = \frac{1.16}{1.1} > 1$$

Separation more effective

Energy Stored in Charged Capacitors

Starting with $U = qV$, Can we say that the energy stored in a capacitor $U = QV$?

No!

Because the charge does not appear on the capacitor instantaneously, the correct equation is $dU = V\,dq$, and we integrate as

$$U = \int_0^Q V\,dq = \int_0^Q \frac{q}{c}\,dq = \frac{1}{2}\frac{Q^2}{C}$$

$$U = \frac{1}{2}\frac{Q^2}{C} \qquad U = \frac{1}{2}QV \qquad U = \frac{1}{2}C\,V^2$$

The energy equations apply for any capacitor structure.

Capacitors with Dielectrics

For any solid or liquid dielectric $\varepsilon = \varepsilon_r \varepsilon_o$, where $\varepsilon_r > 1$.

For example, paper $\varepsilon_r = 3.7$, water $\varepsilon_r = 80$, glass $\varepsilon_r = 5.4$.

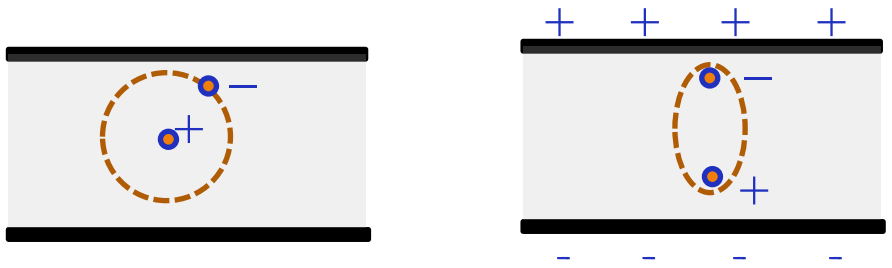
Why do we use dielectric as capacitor filling between conductors?

1) Increase capacitance. $C = \varepsilon_o \varepsilon_r \frac{A}{d}$

2) Increase rigidity of the structure.

3) Increase structure electrical strength (breakdown).

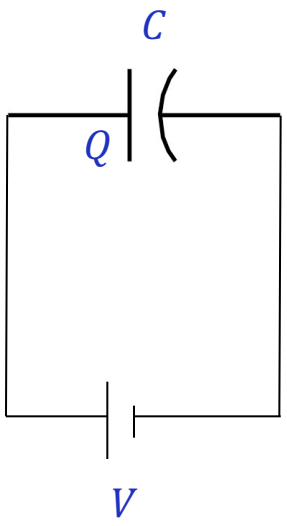
4) The energy stored is less. $U = U_o / \varepsilon_r$ Why?



Material Polarization

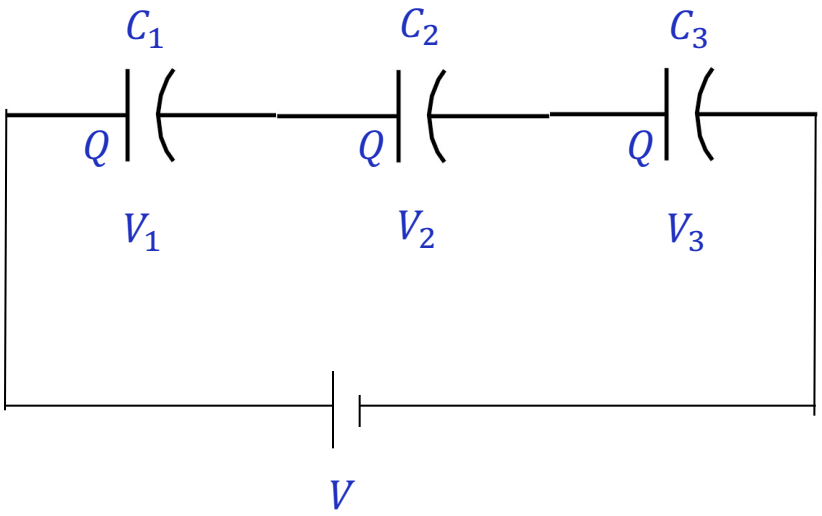
Capacitors in Electrical Circuits

Single



$$C = \frac{Q}{V}$$

Series

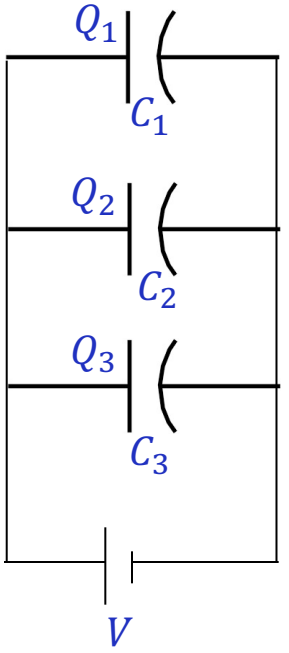


$$V = V_1 + V_2 + V_3$$

$$C_{tot} = \frac{Q}{V_1 + V_2 + V_3}$$

$$C_{tot} = \frac{1}{\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}}$$

Parallel



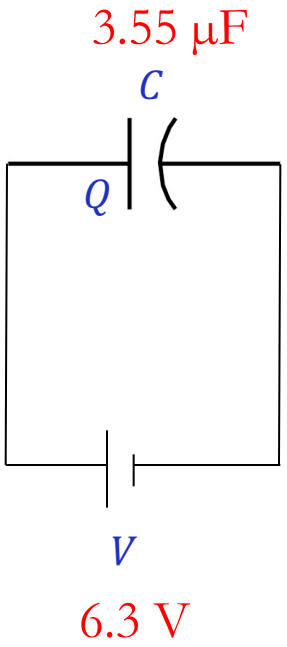
$$Q = Q_1 + Q_2 + Q_3$$

$$C_{tot} = \frac{Q_1 + Q_2 + Q_3}{V}$$

$$C_{tot} = C_1 + C_2 + C_3$$

Capacitors in Electrical Circuits

Example

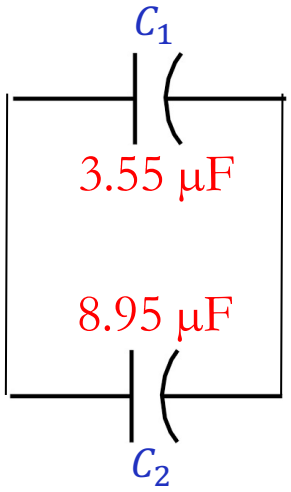


Initial circuit

$$Q = CV$$

$$Q = 22.4\ \mu C$$

Remove battery
Connect
another
capacitor



$$Q_1 = \frac{C_1}{C_1 + C_2} Q = 6.4\ \mu C$$

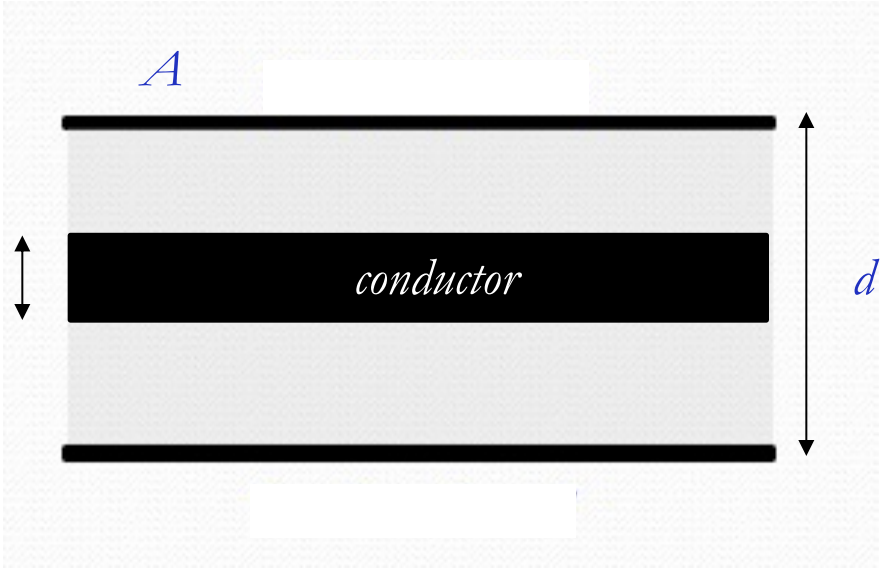
$$V_1 = \frac{Q_1}{C_1} = 1.8\ V$$

$$Q_2 = \frac{C_2}{C_1 + C_2} Q = 16\ \mu C$$

$$V_2 = \frac{Q_2}{C_2} = 1.8\ V$$

Capacitors in Electrical Circuits

Example



$$C_{initial} = \varepsilon \frac{A}{d}$$

Two capacitors in series

$$C_{each} = \varepsilon \frac{A}{(d-a)/2}$$

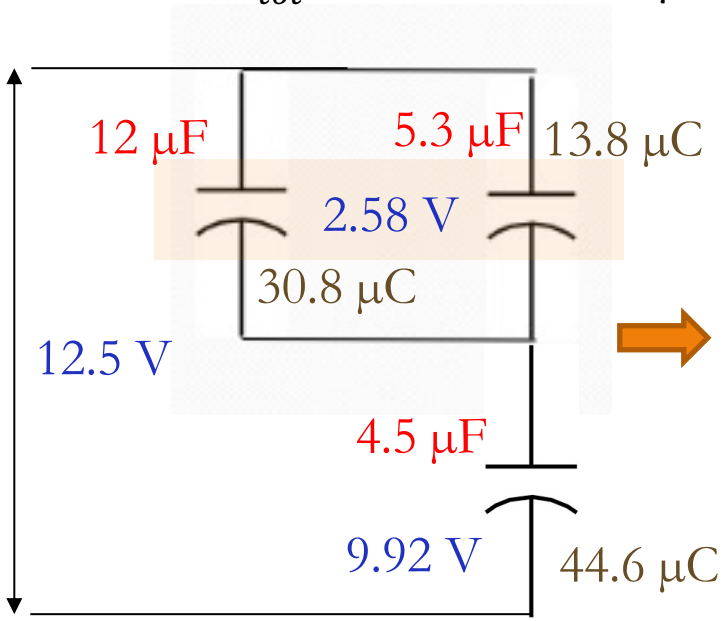
$$C_{tot} = \varepsilon \frac{A}{(d-a)}$$

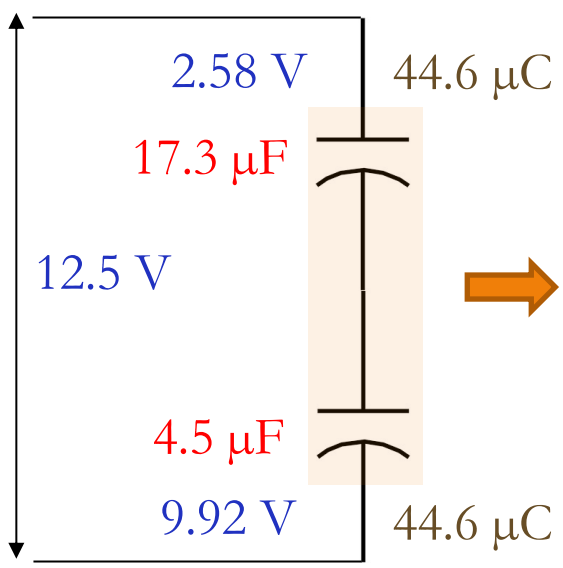
Capacitance increased

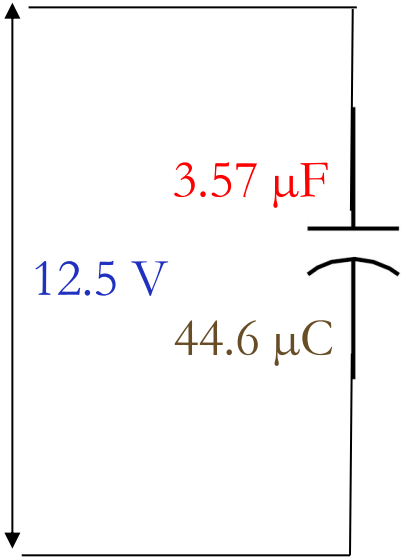
Capacitors in Electrical Circuits

Example: Find charge and potential difference on each capacitor. $Q = C\,V$

$C_{tot} = 12 + 5.3 = 17.3\ \mu F$



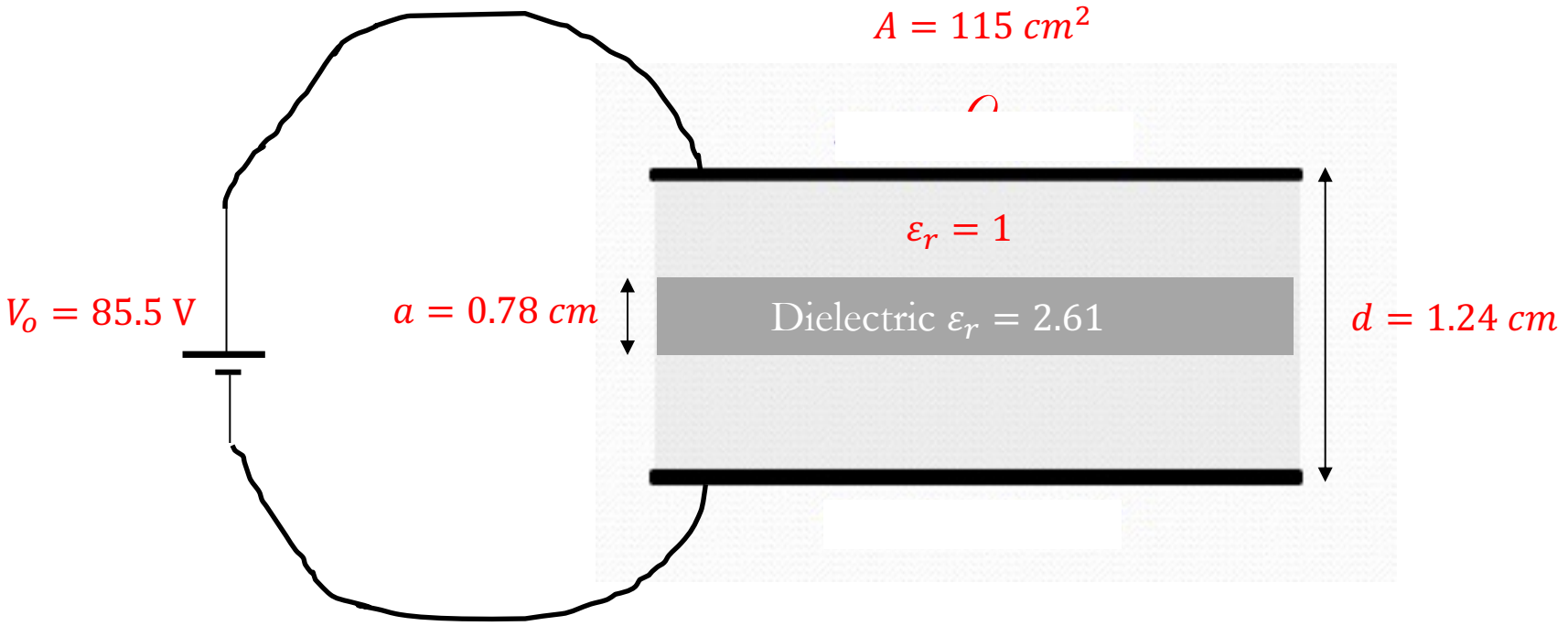




$$C_{tot} = \frac{1}{\frac{1}{17.3} + \frac{1}{4.5}} = 3.57\ \mu F$$

Capacitors in Electrical Circuits

Example: Consider the following capacitor situation.



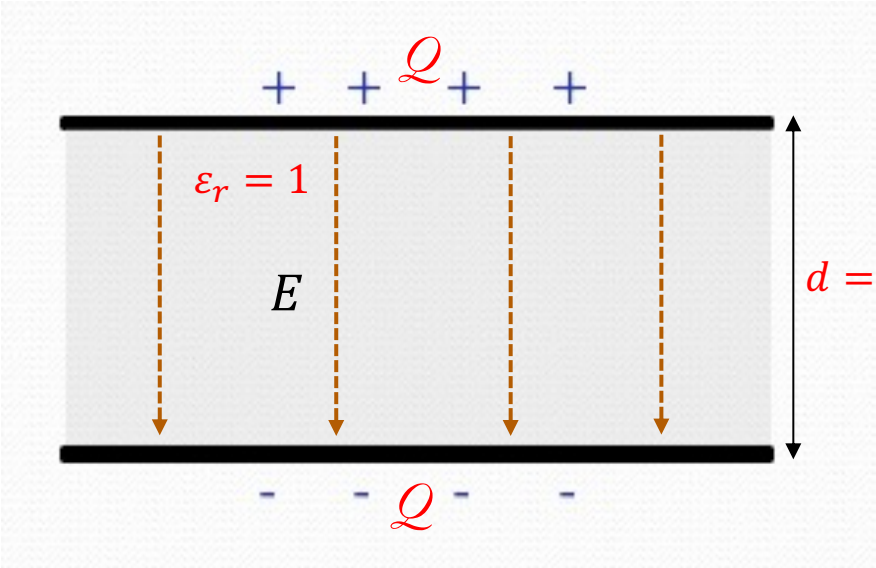
1) Find the initial C, Q, V and E.

2) Find the new C, Q, V and E.

Capacitors in Electrical Circuits

1) Find the initial C, Q, V and E.

$$A = 115\ cm^2$$



$$C_o = \varepsilon_o \frac{A}{d} = \frac{8.85\times10^{-12}\times115\times10^{-4}}{1.24\times10^{-2}}$$

$$C_o = 8.21\ \text{pF}$$

$$V_o = 85.5\ \text{V}$$

$$Q_o = C_o V_o = 8.21\times10^{-12}\times85.5$$

$$Q_o = 702\ \text{pC}$$

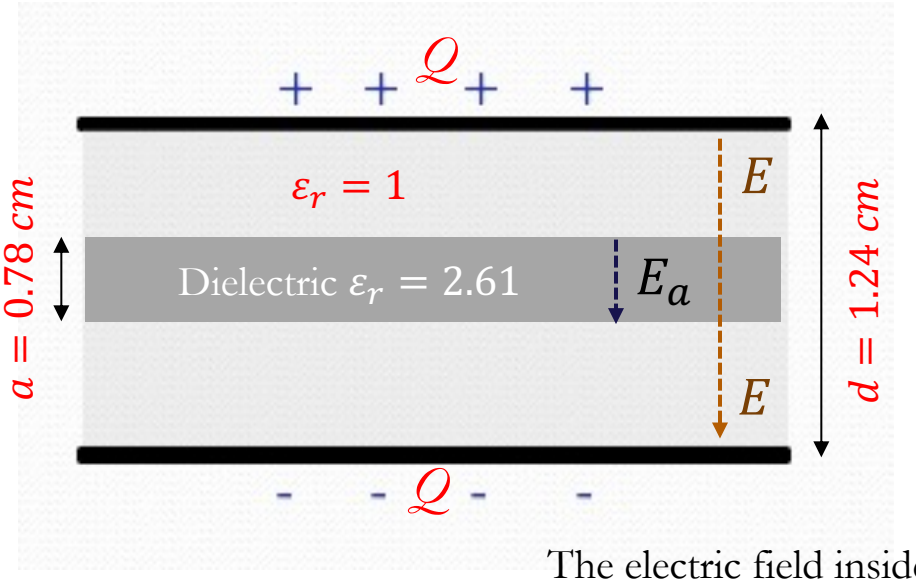
$$E_o = \frac{V_o}{d} = \frac{85.5}{1.24\times10^{-2}}$$

$$E_o = 6.9\ \text{kV/m}$$

Capacitors in Electrical Circuits

2) Find the new C, Q, V and E.

$A = 115\ cm^2$



Charge stays the same! $Q = 702\ \text{pC}$

The electric field inside the dielectric

$$E_a = \frac{E}{\varepsilon_r} = \frac{6.9\times10^3}{2.61} = 2.64\ \text{kV/m}$$

The potential across the capacitor

$$V = -\int_0^d E\ dl\ = E(d-a) + E_a\ a$$

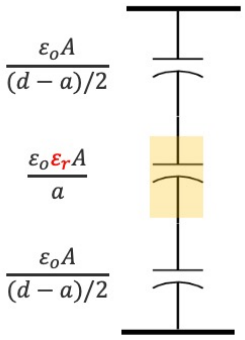
$$V = 6.9\times10^3\times(1.24-0.78)\times10^{-2} + 2.64\times10^3\times0.78\times10^{-2}$$

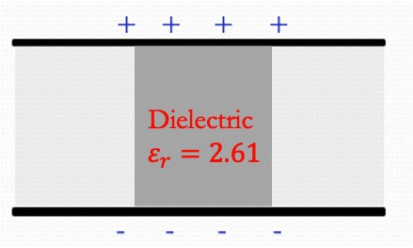
$V = 52.3\ \text{V}$ Why decreased?

$$C = \frac{Q}{V} = \frac{702\times10^{-12}}{52.3}$$

$C = 13.4\ \text{pF}$ Why increased?

The electric field inside the dielectric will be different!

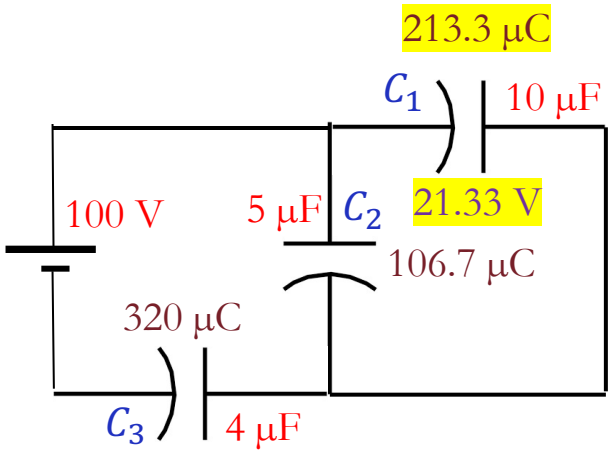


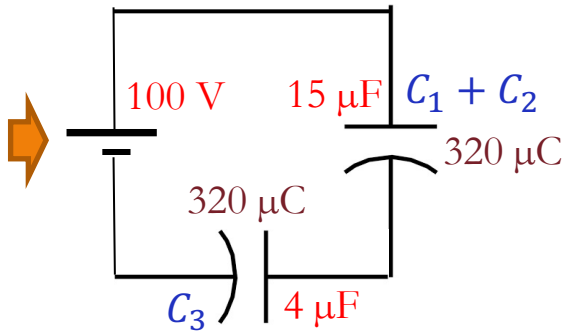


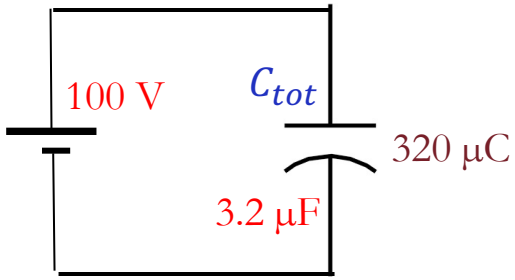
Can you guess the answer here?

Capacitors in Electrical Circuits

Example: After steady-state, C_3 undergoes breakdown (becomes short circuit). Find the change in charge and potential across C_1.

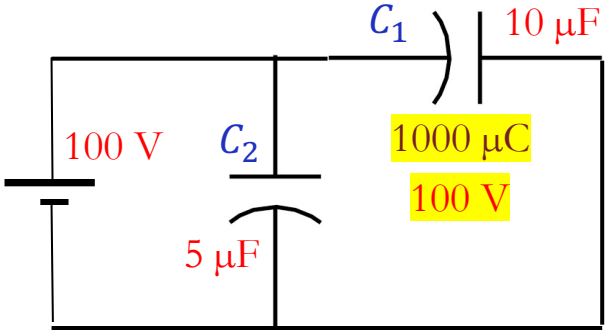






$$C_{tot} = \frac{1}{\frac{1}{C_1 + C_2} + \frac{1}{C_3}}$$

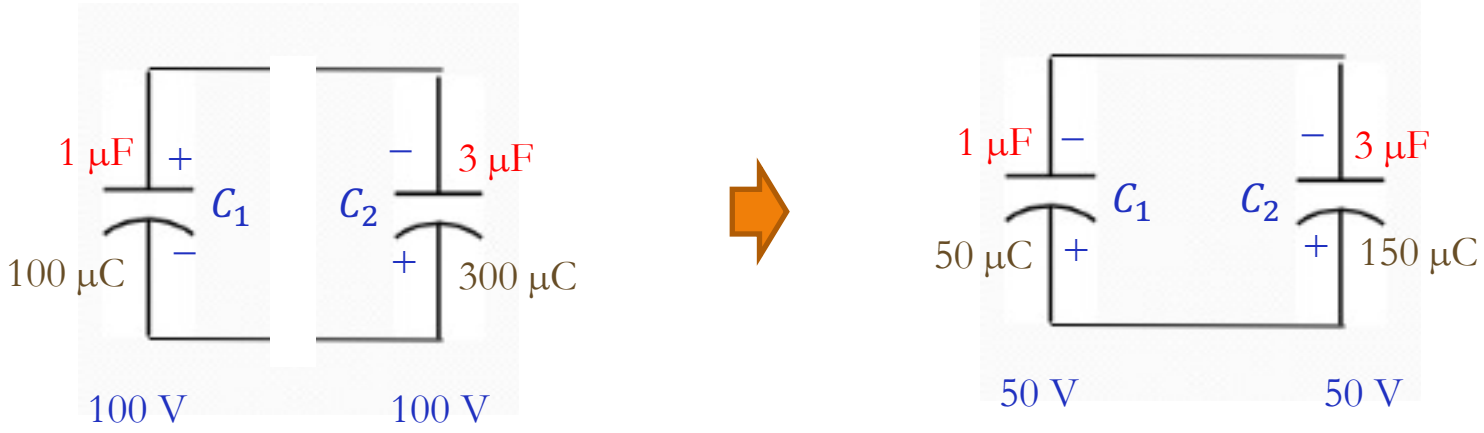
After breakdown



What is the message in this example?

Capacitors in Electrical Circuits

Example: After the two capacitors are connected, find the change in charge and potential across each one.



Total charge is

300 μC -100 μC =200 μC.

It will divide based on capacitance.

Q & A

Send me an email if
you have any
questions.